

HAETAE Line-by-Line EasyCrypt Mathematical Guides

Generated from provable-security/proof-files.txt

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This folder contains one LaTeX guide for each EasyCrypt file in the provable-security manifest. Each guide preserves every source line and gives a mathematical or proof-script reading of that line. The guides are explanatory documents; the checked artifact remains the EasyCrypt files compiled by `provable-security/verify-provable-security.sh`.

EasyCrypt file	Lines	Guide	Role
HAETAE_Security.ec	1272	HAETAE_Security-line-by-line-guide.tex	Top-level correctness and EUF-CMA theorem composition.
Sig_ROM.ec	206	Sig_ROM-line-by-line-guide.tex	Generic signature games and random-oracle interfaces.
HAETAE_HopGames.ec	8817	HAETAE_HopGames-line-by-line-guide.tex	Detailed HAETAE hybrid games, invariants, and game-hop bounds.
HAETAE_Reductions.ec	504	HAETAE_Reductions-line-by-line-guide.tex	Concrete loss terms and genetic normalization.
HAETAE_ROM.ec	111	HAETAE_ROM-line-by-line-guide.tex	Typed random-oracle query and output model.
HAETAE_ROM_Programming.ec	1754	HAETAE_ROM_Programming-line-by-line-guide.tex	Counted random-oracle programming events and bounds.
HAETAE_Scheme.ec	66	HAETAE_Scheme-line-by-line-guide.tex	HAETAE instantiation of the generic signature interface.
HAETAE_Transcript.ec	660	HAETAE_Transcript-line-by-line-guide.tex	Transcript records and validity/extraction facts.
HAETAE_Events.ec	57	HAETAE_Events-line-by-line-guide.tex	Named rejection and acceptance events.
HAETAE_Assumptions.ec	377	HAETAE_Assumptions-line-by-line-guide.tex	Abstract hardness games and reduction interfaces.

EasyCrypt file	Lines	Guide	Role
HAETAE_ Distributions.ec	1218	HAETAE_ Distributions-line-by-line-guide. tex	Sampling distributions, support and algebra point bounds.
HAETAE_Rejection.ec	2204	HAETAE_ Rejection-line-by-line-guide. tex	Rejection predicates and rejection-sampling loss accounting.
HAETAE_Algebra.ec	3885	HAETAE_ Algebra-line-by-line-guide. tex	Deterministic algebraic model and algebra lemmas.
HAETAE_Params.ec	113	HAETAE_ Params-line-by-line-guide. tex	Parameter modes and constants.
HAETAE_FIPS202.ec	105	HAETAE_ FIPS202-line-by-line-guide. tex	Modeled FIPS202/SHAKE support.
HAETAE_Keccak1600. ec	697	HAETAE_ Keccak1600-line-by-line-guide. tex	Keccak state and low-level hash support.

Regenerating

From haetae-security, run:

```
python3 guide/generate-line-by-line-guides.py
```

Building PDFs

From guide/line-by-line, run:

```
make all
# or build one companion:
make HAETAE_Security-line-by-line-guide.pdf
```